

World_insects_estimate

Load libraries and ../Data

```
library(SpadeR)
library(dplyr)
library(stringr)
library(pairwiseCI)
#library(remotes)
#install_version("CARE1", "1.1.0")
library(CARE1)

## Load Data

#Microgastrinae

reared <- read.csv("../Data/Microgastrinae ACG Reared.csv") %>%
  select(BOLD.code, Frequency)

core <- read.csv("../Data/Microgastrinae ACG Core Malaise.csv") %>%
  select(BOLD.code, Frequency)

peri <- read.csv("../Data/Microgastrinae ACG Peripheral Malaise.csv") %>%
  select(BOLD.code, Frequency)

# All insecta core

insec <- read.csv("../Data/Insecta ACG Core Malaise.csv") %>%
  select(BOLD.code, Frequency)

## highlight estimator

estimator <- "      iChao1 (Chiu et al. 2014)"
```

```

## adding in the trees

#TA = number of tree species in ACG
TA <- 1500
TA_alt <- 1200

#Tc = number of tree species in CR

TC <- 2500
#Tw = number of tree species on Earth

## estimated tree species

TW <- 73274
TW_CI <- c(73271, 73276)

TW_2 <- 89147

## described number of tree species

TW_d <- 60065

### Number of Saturniid moths

SA <- 106
SW <- 4535

```

Approach 1: Calculate insect richness worldwide using the CARE estimator for Microgastrinae

```

### Total number of BINS

reared %>% nrow()

```

```
[1] 886
```

```
core %>% nrow()
```

```
[1] 388
```

```
peri %>% nrow()
```

```
[1] 576
```

```
###
```

Step 1: Combining Malaise and Reared samples of Microgastrinae and estimating the true richness of ACG Microgastrinae

```
full_micro_table <- reared %>%  
  rename(reared = Frequency) %>%  
  full_join(rename(core, core = Frequency)) %>%  
  full_join(rename(peri, peri = Frequency)) %>%  
  mutate(reared = ifelse(is.na(reared), 0, 1)) %>%  
  mutate(core = ifelse(is.na(core), 0, 1)) %>%  
  mutate(peri = ifelse(is.na(peri), 0, 1))
```

```
Joining with `by = join_by(BOLD.code)`
```

```
Joining with `by = join_by(BOLD.code)`
```

```
head(full_micro_table)
```

	BOLD.code	reared	core	peri
1	BOLD:AAA1266	1	0	0
2	BOLD:AAA1634	1	1	1
3	BOLD:AAA2143	1	0	0
4	BOLD:AAA2511	1	0	0
5	BOLD:AAA2668	1	0	0
6	BOLD:AAA3229	1	0	1

```
nrow(full_micro_table)
```

```
[1] 1411
```

```
# Transform observed ../Data to CARE1 input format  
micro_care <- as.record(full_micro_table[, -1])  
  
micro_care
```

```
001 010 011 100 101 110 111
246 133 146 663 114 39 70
```

```
# Main step to obtain population size estimates
CARE1.print(micro_care)
```

```
Warning in ni/nhat: Recycling array of length 1 in vector-array arithmetic is deprecated.
  Use c() or as.vector() instead.
Warning in ni/nhat: Recycling array of length 1 in vector-array arithmetic is deprecated.
  Use c() or as.vector() instead.
Warning in ni/nhat: Recycling array of length 1 in vector-array arithmetic is deprecated.
  Use c() or as.vector() instead.
```

(1) NUMBER OF IDENTIFIED CASES IN EACH LIST:

```
n1 n2 n3
886 388 576
```

(2) ESTIMATES BASED ON ANY PAIR OF SAMPLES:

	Petersen	Chapman	se	cil	ciu
pa12	3154	3136	236	2725	3655
pa13	2774	2765	149	2501	3087
pa23	1035	1033	37	970	1115

Note1: Refer to Seber(1982,pages 59 and 60) for Petersen estimator and Chapman estimators as well as s.e formula.

Note2: A log-transformation is used is used to obtain the confidence interval so that the lower limit is always greater than the number of ascertained. Refer to Chao(1987,Biometrics,43,783-791) for the construction of the confidence interval.

(3) SAMPLE COVERAGE APPROACH:

	M	D	Chat	est	se	cil	ciu
Nhat-0	1411	1063.667	0.494	2153	61	2043	2283
Nhat	1411	1063.667	0.494	3480	509	2697	4740
Nhat-1	1411	1063.667	0.494	2402	105	2217	2630

Warning: The estimated sample coverage(overlapping information)is too low so that Nhat is unstable. Recommend the use of Nhat-0 or Nhat-1.

Parameter estimates:

	u1	u2	u3	r12	r13	r23
Nhat-0	0.41	0.18	0.27	-0.32	-0.22	1.08

```
Nhat    0.25 0.11 0.17  0.10  0.25 2.36
Nhat-1  0.37 0.16 0.24 -0.24 -0.13 1.32
```

Definitions for the sample coverage approach:

M: number of individuals ascertained in at least one list.

D: the average of the number of individuals listed in the combination of any two lists omitting the other one.

$\hat{C}$: sample coverage estimate, see Equation (14) of Chao and Tsay(1998).
est: population size estimate.

se: estimated standard error of the population size estimation based on bootstrap replications.

cil: 95% confidence interval lower limit(using a log-transformation).

ciu: 95% confidence interval upper limit(using a log-transformation).

Nhat: Population size estimate for sufficiently high sample coverage cases, see Equation (20) of Chao and Tsay (1998).

Nhat-1: One-step population size estimate for low sample coverage cases; see Equation (2.21) of Chao et al. (1996). This estimator is suggested for use when the estimated se of Nhat is relatively large.

u1,u2,u3: estimated mean probabilities depending on the estimate of N.

r12,r13,r23 etc.: estimated coefficient of covariation(CCV) depending on the estimate of N.

```
# Print the CARE1 input format

## n1 = reared = 886
## n2 = core = 388
## n3 = peripheral = 576

## Using Nhat-1

Nhat1 <- round(estN(micro_care, method = "LSC", se = TRUE, nboot = 200))

micro_all_estimate <- Nhat1[1]
micro_all_CI <- c(Nhat1[3], Nhat1[4])
```

Step 2. Estimating the undersampling ratio for Core Microgastrinae.

```
## proportion of core malaise of the ichao total
```

```
proportion_core <- nrow(core)/micro_all_estimate
```

```
proportion_core
```

```
[1] 0.1615321
```

```
CI_proportion_core <- nrow(core)/micro_all_CI
```

```
CI_proportion_core
```

```
[1] 0.1740691 0.1485452
```

Step 3 Calculate total number of insects in ACG

```
## Total number of insecta bins in ACG core malaise
```

```
n_insects_core <- nrow(incsc)
```

```
n_insects_core
```

```
[1] 53945
```

```
total_insec_acg <- n_insects_core*(1/proportion_core)
```

```
total_insec_acg
```

```
[1] 333958.5
```

```
CI_total_insect_acg <- n_insects_core*(1/round(CI_proportion_core,3))
```

```
n_insects_core*(1/round(CI_proportion_core,3))
```

```
[1] 310028.7 362047.0
```

```
n_insects_core*(1/CI_proportion_core)
```

```
[1] 309905.7 363155.5
```

```
CI_total_insect_acg
```

```
[1] 310028.7 362047.0
```

Step 4. Estimating global insect richness – based on trees

```
##### Total insects
```

```
print("Total world insects based on estimated tree richness")
```

```
[1] "Total world insects based on estimated tree richness"
```

```
total_insect_world <- (TW*total_insec_acg)/TA  
total_insect_world
```

```
[1] 16313649
```

```
CI_total_insect_world <- (TW*CI_total_insect_acg)/TA  
CI_total_insect_world
```

```
[1] 15144697 17685754
```

```
print("Total world insects based on described trees")
```

```
[1] "Total world insects based on described trees"
```

```
total_insect_world <- (TW_d*total_insec_acg)/TA  
total_insect_world
```

```
[1] 13372811
```

```
CI_total_insect_world <- (TW_d*CI_total_insect_acg)/TA  
CI_total_insect_world
```

```
[1] 12414584 14497568
```

```
##### Total insects
```

```
print("Total world insects based on estimated tree richness")
```

```
[1] "Total world insects based on estimated tree richness"
```

```
total_insect_world <- (TW*total_insec_acg)/TA_alt  
total_insect_world
```

```
[1] 20392061
```

```
CI_total_insect_world <- (TW*CI_total_insect_acg)/TA_alt  
CI_total_insect_world
```

```
[1] 18930871 22107192
```

```
print("Total world insects based on described trees")
```

```
[1] "Total world insects based on described trees"
```

```
total_insect_world <- (TW_d*total_insec_acg)/TA_alt  
total_insect_world
```

```
[1] 16716013
```

```
CI_total_insect_world <- (TW_d*CI_total_insect_acg)/TA_alt  
CI_total_insect_world
```

```
[1] 15518230 18121960
```

Step 4. Estimating global insect richness – based on saturniid moths


```
##### Total insects
```

```
print("Total world insects based on Saturniid moths")
```

```
[1] "Total world insects based on Saturniid moths"
```

```
total_insect_world <- (SW*total_insec_acg)/SA  
total_insect_world
```

```
[1] 14287752
```

```
CI_total_insect_world <- (SW*CI_total_insect_acg)/SA  
CI_total_insect_world
```

```
[1] 13263965 15489463
```

An alternative estimate

```
##### Total insects
```

```
print("Total world insects based on estimated tree richness")
```

```
[1] "Total world insects based on estimated tree richness"
```

```
total_insect_world <- (TW_2*total_insec_acg)/TA  
total_insect_world
```

```
[1] 19847598
```

```
CI_total_insect_world <- (TW_2*CI_total_insect_acg)/TA  
CI_total_insect_world
```

```
[1] 18425421 21516935
```

```
##### Total insects
```

```
print("Total world insects based on estimated tree richness")
```

```
[1] "Total world insects based on estimated tree richness"
```

```
total_insect_world <- (TW_2*total_insec_acg)/TA_alt  
total_insect_world
```

```
[1] 24809497
```

```
CI_total_insect_world <- (TW_2*CI_total_insect_acg)/TA_alt  
CI_total_insect_world
```

```
[1] 23031776 26896168
```